

Know It’s Absent, Yet Point Anyway: Cross-Task Semantic Contradictions in Vision-Language Models

Anonymous ACL submission

Abstract

Vision-language models (VLMs) increasingly support both visual question answering and spatial grounding, yet their behavior across these tasks can be inconsistent. In this paper, we identify **Existence–Localization Contradiction**, a failure mode where a VLM correctly recognizes that a queried object is absent from an image, but still produces a hallucinated localization when asked to ground the same referent. We introduce a conditional evaluation protocol that pairs existence questions with downstream localization prompts over the same image and target expression, isolating cases where models know the referent is absent but fail to carry this judgment into grounding. Experiments on strong VLMs show that such contradictions are common, revealing a gap between perception, abstention, and localization. We further show that lightweight interventions can substantially reduce these failures. Our findings highlight that reliable VLM grounding requires not only accurate recognition, but also consistent refusal to localize absent referents.

1 Introduction

Vision-language models (VLMs) (Liu et al., 2024; Team et al., 2025; Zhu et al., 2025; Wang et al., 2025; Bai et al., 2025b,a) can now follow natural-language instructions over images and increasingly support spatial outputs such as points and bounding boxes. This makes them useful not only for visual question answering, but also as language-conditioned grounding modules in downstream systems. As a result, their reliability matters not only at the answer level, but also at the level of spatial commitments.

Prior work has shown that VLMs often hallucinate entities when language presupposes their existence. Antidote (Wu et al., 2025) studies counterfactual presupposition questions, JBA (Li et al., 2025) evaluates false-premise recognition,

and Ground-V (Zong et al., 2025) considers hallucinated references in complex grounding settings. However, these settings do not directly test whether a model that has already recognized object absence can preserve that judgment when later asked to localize the same referent.

We study this gap through a simple but important failure case. A VLM may correctly answer “No” to the question “Is there a cat in the image?”, yet still return a bounding box when later asked “Point to the cat.” We define this failure as **Existence–Localization Contradiction**.

Unlike standard no-target grounding or false-premise recognition errors, Existence–Localization Contradiction captures whether a model preserves an absence judgment under a downstream localization request. The model semantically rejects an entity, yet operationally commits to it through localization output.

This semantic contradiction not only highlights a violation of logical consistency between a model’s semantic belief and its subsequent action commitment, but also exposes a critical risk in multimodal systems where localization serves as actionable grounding evidence. A hallucinated bounding box for a nonexistent referent can propagate false evidence and trigger incorrect downstream actions. We therefore characterize hallucinated localization as a form of *actionable hallucination*. Such failures are particularly critical in robotics, medical localization, autonomous agents, retrieval pipelines, and grounding-based tool use, where localization outputs directly influence system behavior. Reliable grounding thus requires not only accurate perception, but also consistent transfer of absence information across task formats.

To measure this behavior, we propose a conditional evaluation protocol. For each image and target expression, we first ask an existence question, then evaluate localization only when the model has correctly judged the target to be absent. This iso-

lates the contradiction of interest: the model knows the referent is absent, yet still produces a non-null spatial output.

We use this setting to answer two questions: (1) how often do VLMs change from a correct absence judgment to an incorrect non-null localization, (2) how can these contradictions be reduced without harming localization performance when the referent is present? To investigate the mechanism behind this failure, we analyze model hidden states under existence and localization prompts, and we use these observations to motivate a lightweight inference-time intervention. Our results suggest that reliable VLM grounding requires not only recognizing when an object is absent, but also carrying that judgment into downstream localization decisions.

2 Related Work

Hallucination in VLMs. Hallucination in vision-language models (VLMs) has been widely studied across several related settings. First, object-level hallucination evaluates whether models incorrectly affirm objects that are not present in the image. POPE (Li et al., 2023b) shows that large VLMs often hallucinate objects, especially when they are plausible given the scene context or co-occurring categories. More recent work studies a stronger object-absence setting by explicitly removing objects from images and testing whether models still “see” the removed entities (He et al., 2025). Second, no-target grounding studies cases where a referring expression has no valid visual referent. RECENTFormer (Hemanthage et al., 2024) investigates referring expression comprehension under varying numbers of valid referents, including no-target cases, while GREx (Ding et al., 2026) generalizes referring expression comprehension, segmentation, and generation to single-target, multi-target, and no-target settings. Similarly, (Yin et al., 2025) propose the ROD dataset to help VLMs reject non-existent objects under grounding queries effectively. Third, false-premise and counterfactual-presupposition work examines whether multimodal models can reject invalid assumptions expressed in language. Antidote (Wu et al., 2025) highlights VLM failures on counterfactual presupposition questions, and JBA (Li et al., 2025) further studies false-premise recognition in multimodal question answering. Together, these works show that absent or invalid referents remain a persistent

challenge for VLMs. However, they mainly focus on existence recognition, premise rejection, or whether the final grounding output is null. Our work instead asks whether a model that has already correctly denied an object’s existence will nevertheless produce a non-null localization for that same object.

Consistency in VLMs. Consistency has recently emerged as an important dimension of VLM robustness and trustworthiness. MM-R3 shows that models can achieve competitive accuracy while still exhibiting substantial inconsistency under rephrasing, restyling, and context perturbations (Chou et al., 2025). Similar observations have been made in broader work on multimodal consistency and robustness (Zhang et al., 2024). Our work shares this motivation, but studies a narrower form of inconsistency: a contradiction between two outputs for the same image and the same queried referent, namely an existence judgment and a localization commitment.

Activation steering. Many hallucination mitigation methods reduce unsupported outputs through contrastive decoding, decoding-time penalties, or post-hoc correction (Leng et al., 2023; Huang et al., 2024; Zhou et al., 2024; Yin et al., 2024). Orthogonal to these approaches, activation steering methods intervene directly in hidden representation space without updating model parameters. A common construction is the difference-in-means estimator, where a steering direction is computed from the mean activation of desired examples minus that of undesired examples. This idea appears in Activation Addition and Contrastive Activation Addition (Turner et al., 2024; Rinsky et al., 2024), as well as in truthfulness and safety interventions that identify linear directions associated with truthful responses or refusal behavior (Li et al., 2023a; Marks and Tegmark, 2024; Ardit et al., 2024). Recent VLM work further suggests that hallucination-related behaviors are encoded in intermediate hidden states and can be reduced through inference-time intervention (Su et al., 2025). Building on this line of work, we construct a DiffMean steering vector from contrastive hallucinated and visually grounded activations, and apply it at inference time to encourage absence-consistent localization.



Figure 1: An example of Existence–Localization Contradiction.

3 Existence–Localization Contradiction

3.1 Problem Definition

We study a failure mode that we call **Existence–Localization Contradiction**. Given an image v and an object query p , we consider two queries: (1) an *existence query* that asks whether object o is present in the image, and (2) a *localization query* that asks the model to point to, or otherwise localize, object o in the same image. For example, in Figure 1, the top prompt asks about the existence of a person, while the bottom prompt asks for the exact position of a person.

In this paper, we focus exclusively on **object absence**, i.e., a setting in which the queried object does not appear in the image at all. A trustworthy model should satisfy two conditions simultaneously:

1. it should correctly deny the presence of the queried object at the existence stage; and
2. it should abstain from localization, or return a null localization, at the localization stage.

We say that a model exhibits an **Existence–Localization Contradiction** if it (1) correctly predicts that the queried object is absent, yet (2) still produces a non-null localization for that same object. Figure 1 demonstrates the phenomenon: under the object existence setting, the model correctly recognizes that no person is present in the image. However, in the object location setting, the model still predicts the location of a person. This distinction is important. A model may fail because it does not recognize object absence in the first place; that is, an object recognition or perception failure. By contrast, the contradiction we study arises when

the model *already knows the object is absent* but still commits to a spatial prediction.

3.2 Conditional Evaluation Protocol

We evaluate the existence–localization contradiction using a two-stage conditional protocol. The goal of this protocol is to test whether a model that correctly recognizes the absence of a queried referent can preserve this judgment when subsequently asked to produce a spatial grounding output.

Let

$$\mathcal{D} = \{(v_i, o_i, e_i)\}_{i=1}^N$$

denote a set of image–object pairs, where v_i is an image, o_i is a queried object, and $e_i \in \{0, 1\}$ is the ground-truth existence label. We define $e_i = 1$ if o_i is present in v_i , and $e_i = 0$ otherwise. Since our failure mode concerns object absence, we focus on the negative subset:

$$\mathcal{N} = \{i : e_i = 0\}.$$

For each pair, we query the same model with two task formats: an existence prompt and a localization prompt. The existence prompt asks whether the queried object is present in the image, while the localization prompt asks the model to return a spatial grounding output for the same queried object.

Stage 1: Existence Recognition. For each image–object pair (v_i, o_i) , we first issue an existence query using a template T_E :

$$q_i^E = T_E(o_i).$$

The model response q_i^E is parsed into a binary existence prediction

$$\hat{e}_i \in \{0, 1\},$$

where $\hat{e}_i = 1$ indicates that the model predicts the queried object as present, and $\hat{e}_i = 0$ indicates that the model predicts it as absent.

This stage evaluates whether the model can explicitly determine the existence status of the queried object before any localization request is considered. For negative examples $i \in \mathcal{N}$, the model correctly recognizes absence when:

$$\hat{e}_i = 0.$$

Stage 2: Localization under Absence. We then query the same model on the same image–object pair with a localization template T_L :

$$q_i^L = T_L(o_i).$$

The localization response q_i^L is parsed into a binary localization decision

$$\hat{\ell}_i \in \{0, 1\},$$

where $\hat{\ell}_i = 1$ indicates that the model returns a non-null spatial prediction, and $\hat{\ell}_i = 0$ indicates abstention or a null localization.

In our experiments, a non-null localization corresponds to a predicted bounding box. If the response contains a parseable bounding box, we set $\hat{\ell}_i = 1$. If the response contains no parseable bounding box, or explicitly states that the queried object is absent and therefore cannot be localized, we set $\hat{\ell}_i = 0$.

3.3 Evaluation Metrics

We report three metrics: Existence Accuracy, Null Localization Precision, and Conditional Existence–Localization Faithfulness.

Existence Accuracy. This metric measures whether the model correctly predicts the presence or absence of the queried object, evaluating the model’s original object-level recognition ability:

$$\text{EA} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{e}_i = e_i].$$

Null Localization Precision. A simple but incorrect way to mitigate the Existence–Localization Contradiction is to return *null localization*—output no bounding box—regardless of the object’s presence. To measure the reliability of the model decision, we additionally define Null Localization Precision as the fraction of localization rejection that occurs on truly negative examples:

$$\text{NLP} = \frac{\sum_{i=1}^N \mathbf{1}[e_i = 0] \mathbf{1}[\hat{\ell}_i = 0]}{\sum_{i=1}^N \mathbf{1}[\hat{\ell}_i = 0]}.$$

A higher NLP indicates that if the model does not provide localization, the object is more likely to be absent. When the denominator is zero, i.e., when the model never abstains from localization queries, we define $\text{NLP} = 0$.

Conditional Existence–Localization Faithfulness. Our primary metric is Conditional Existence–Localization Faithfulness (CELF), which measures whether the model preserves its correct absence judgment when asked to localize the same queried object. Specifically, CELF measures how often the model returns a null localization given that it correctly recognizes the object’s absence under the existence prompt:

$$\text{CELF} = \frac{\sum_{i \in \mathcal{N}} \mathbf{1}[\hat{e}_i = 0] \mathbf{1}[\hat{\ell}_i = 0]}{\sum_{i \in \mathcal{N}} \mathbf{1}[\hat{e}_i = 0]}.$$

Equivalently, CELF estimates:

$$\Pr(\hat{\ell}_i = 0 \mid e_i = 0, \hat{e}_i = 0).$$

A higher CELF indicates stronger existence–localization faithfulness. In particular, $\text{CELF} = 1$ means that whenever the model correctly predicts the queried object as absent, it also refuses to localize it. Conversely, $\text{CELF} = 0$ means that even when the model correctly recognizes absence, it still always returns a non-null localization.

When the denominator is zero, i.e., when a model never correctly predicts absence on the negative subset, CELF is undefined and excluded from aggregate comparisons.

Relation to Standard Evaluations This protocol differs from two related evaluation settings. First, it differs from false-premise recognition because we do not only test whether the model can reject an invalid linguistic presupposition. Instead, we test whether the model preserves this rejection when the task format changes from existence recognition to localization. A model may correctly answer that an object is absent, yet still produce a bounding box when later asked to localize the same object. Second, it differs from standard no-target grounding evaluation. CELF is conditioned on cases where the model has already correctly recognized absence. This conditioning isolates a stricter form of consistency: whether the model’s localization behavior remains faithful to its own prior absence judgment.

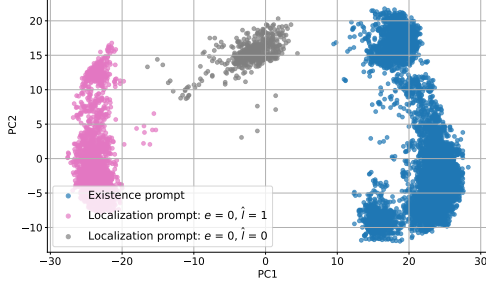


Figure 2: PCA projection of hidden states.

4 Mitigating Existence–Localization Contradiction by Light-weight Steering

4.1 Hidden Space Analysis

To better understand the model’s behavior, we analyze the feature space on different scenarios. For each sample, we collect the hidden states at the final token position under the existence prompt and the localization prompt. We project those vectors into a two-dimensional space using PCA and visualize the hidden states of Qwen2.5-VL-7B-Instruct in Figure 2.

First, we observe that the hidden states on existence prompts form a distinct cluster than the hidden states on localization prompts. This separation suggests that the model enters different representational regimes when solving those two tasks and might not utilize well the benefits of multi-task learning. We further zoom into the feature space of localization prompts and inspect two cases: samples where the model incorrectly returns the bounding-box for non-existence object ($e = 0, \hat{l} = 1$), and samples where the model correctly realizes the object’s absence ($e = 0, \hat{l} = 0$). Figure 2 demonstrates that these two groups are also separable and form two distinguishable regions in the feature space, implying that the model internally encodes the signal that refuses to answer the localization prompt. More importantly, the region of correct null-localization responses is closer to the region of responses to existence prompts. This observation confirms our hypothesis: Existence–Localization Contradiction happens due to the lack of sharing knowledge in multi-task learning; in the case of the correct null-localization, the model utilizes information about object’s existence.

The Existence–Localization Contradiction arises from limited knowledge sharing between existence prediction and object localization.

4.2 Lightweight Steering-Based Mitigation

Section 4.1 suggests that nudging the hidden states towards the existence prediction region allows the model to leverage information from the existence prediction task and correctly realize the object’s absence. In this paper, we propose a lightweight steering-based intervention to mitigate object-localization hallucinations. Our method follows the general latent-space steering paradigm of ASD (Su et al., 2025), which estimates an intervention direction from contrastive paired outputs and applies that direction at test time to shift model representations toward more faithful behavior.

We construct a paired intervention set

$$\mathcal{P} = \{(v_i, o_i, q_i^L, y_i^{\text{loc}}, y_i^{\text{abs}})\}_{i=1}^M,$$

where v_i denotes an image, o_i denotes the queried object, and q_i^L is the localization query. For each query, y_i^{loc} is a response that yields a predicted bounding-box, whereas y_i^{abs} is an response that explicitly states that the queried object does not exist. The pair $(y_i^{\text{loc}}, y_i^{\text{abs}})$ is designed to capture the contrast between hallucination-prone localization behavior and evidence-consistent abstention behavior under the same image–object query. This choice is also consistent with our evaluation protocol, in which the desired behavior on object-absence cases is null localization rather than non-null spatial grounding.

We estimate the steering direction from hidden-state differences between contrastive paired outputs. For each pair, we run the model on both responses and extract the hidden representation at transformer layer l . Let

$$\mathbf{h}_{i,\text{loc}}^{(l)} \quad \text{and} \quad \mathbf{h}_{i,\text{abs}}^{(l)}$$

denote the hidden states associated with the localization and abstention responses, respectively. We then compute a contrastive difference vector

$$\Delta \mathbf{h}_i^{(l)} = \mathbf{h}_{i,\text{abs}}^{(l)} - \mathbf{h}_{i,\text{loc}}^{(l)}.$$

Intuitively, $\Delta \mathbf{h}_i^{(l)}$ captures the latent shift from non-faithful localization behavior toward abstention-consistent behavior. Given the set of

Table 1: Performance evaluation of our method against baselines.

Model	Type	POPE			AMBER			MME (object existence)			DASH-B		
		EA ↑	NLP ↑	CELF ↑	EA ↑	NLP ↑	CELF ↑	EA ↑	NLP ↑	CELF ↑	EA ↑	NLP ↑	CELF ↑
Qwen2.5-VL-7B-Instruct	Base	83.7	94.7	10.4	95.4	100.0	28.6	100.0	100.0	20.0	74.3	100.0	2.8
	Steered	82.2 _{-1.5}	94.8 _{+0.1}	41.6 _{+31.2}	97.2 _{+1.8}	100.0 _{+0.0}	53.1 _{+24.5}	95.0 _{-5.0}	100.0 _{+0.0}	90.0 _{+70.0}	76.4 _{+2.1}	95.0 _{-5.0}	13.5 _{+10.7}
Qwen3-VL-8B-Instruct	Base	89.4	95.2	15.8	90.2	100.0	38.2	98.3	100.0	50.0	74.5	100.0	2.7
	Steered	88.4 _{-1.0}	96.1 _{+0.9}	50.0 _{+34.2}	92.7 _{+2.5}	100.0 _{+0.0}	65.3 _{+27.1}	98.3 _{+0.0}	100.0 _{+0.0}	80.0 _{+30.0}	75.7 _{+1.2}	92.3 _{-7.7}	13.1 _{+10.4}
InternVL3-8B	Base	90.8	0.0	0.0	92.3	0.0	0.0	98.3	0.0	0.0	67.6	0.0	0.0
	Steered	90.4 _{-0.4}	84.6 _{+84.6}	72.9 _{+72.9}	92.6 _{+0.3}	100.0 _{+100.0}	55.0 _{+55.0}	98.3 _{+0.0}	79.3 _{+79.3}	76.7 _{+76.7}	69.8 _{+2.2}	68.9 _{+68.9}	53.3 _{+53.3}
InternVL3.5-8B	Base	86.7	0.0	0.0	88.5	0.0	0.0	100.0	0.0	0.0	68.9	0.0	0.0
	Steered	85.2 _{-1.5}	46.0 _{+46.0}	20.6 _{+20.6}	88.2 _{-0.3}	100.0 _{+100.0}	33.4 _{+33.4}	100.0 _{+0.0}	51.9 _{+51.9}	46.7 _{+46.7}	71.2 _{+2.3}	33.9 _{+33.9}	15.6 _{+15.6}

difference vectors $\{\Delta \mathbf{h}_i^{(l)}\}_{i=1}^M$, we compute the steering vector as the mean of difference across the dataset

$$\mathbf{s}^{(l)} = \frac{1}{M} \sum_{i=1}^M \Delta \mathbf{h}_i^{(l)}$$

At inference time, we inject the steering direction additively into the hidden state at layer l :

$$\tilde{\mathbf{h}}^{(l)} = \mathbf{h}^{(l)} + \alpha \mathbf{s}^{(l)},$$

where $\alpha \in \mathbb{R}$ controls the intervention strength. Our method performs the intervention at only one transformer layer. This makes the intervention parameter-free, localized, and inference-time efficient, while still allowing the model to be steered away from contradiction-prone representations.

Since the effectiveness of steering depends on both the intervention location and the intervention strength, we perform a grid search over the target layer l and the scaling coefficient α . The final configuration is selected based on validation performance, with priority given to improving Conditional Existence–Localization Faithfulness (CELF) and reducing contradiction cases, while preserving the model’s utility on standard localization examples. This objective directly matches our conditional evaluation framework, where the central goal is not merely to suppress localization outputs in general, but to ensure that localization behavior remains faithful to the model’s own correct absence judgment.

5 Experimental Results

5.1 Implementation Details

We conduct experiments on four vision-language models, namely Qwen2.5-VL-7B-Instruct (Bai et al., 2025b), Qwen3-VL-8B-Instruct, InternVL3-8B (Zhu et al., 2025), and InternVL3.5-8B (Wang et al., 2025). We evaluate these models on four

object-existence and hallucination-related benchmarks: POPE (Li et al., 2023b), AMBER on the object existence subset (Wang et al., 2024), MME on the object existence subset (Fu et al., 2026), and DASH-B (Augustin et al., 2025). For each model, we construct a paired set of 100 examples to estimate the steering direction. We use EasySteer (Xu et al., 2025) to create steer vectors. We then apply the steering vector at inference time through an additive intervention on the hidden state. For each model, we perform a grid search over the intervention layer l and the scaling coefficient α in order to identify the most effective steering configuration. The full search space and the selected values of α and layer for each model are provided in Appendix A. In term of prompt settings, for existence recognition, we use the original prompts provided by each benchmark. For localization, we design model-specific prompt templates, since different models require different prompting formats to reliably produce spatial grounding outputs. The prompt can be found in Appendix B. We then evaluate each model under the conditional protocol, and report Existence Accuracy (EA), Null Localization Precision (NLP), and Conditional Existence–Localization Faithfulness (CELF).

5.2 VLMs Suffer from Existence–Localization Contradiction

Table 1 reveals a consistent pattern across models and datasets. Although all models achieve relatively strong existence recognition performance, this ability does not transfer reliably to the downstream localization setting. In particular, EA is generally high across benchmarks, reaching above 90.0 on several settings and even 100 on MME for Qwen2.5-VL-7B-Instruct and InternVL3.5-8B-HF. However, both NLP and CELF remain strikingly low in most cases, indicating that the models frequently produce non-null localization outputs even when they should abstain.

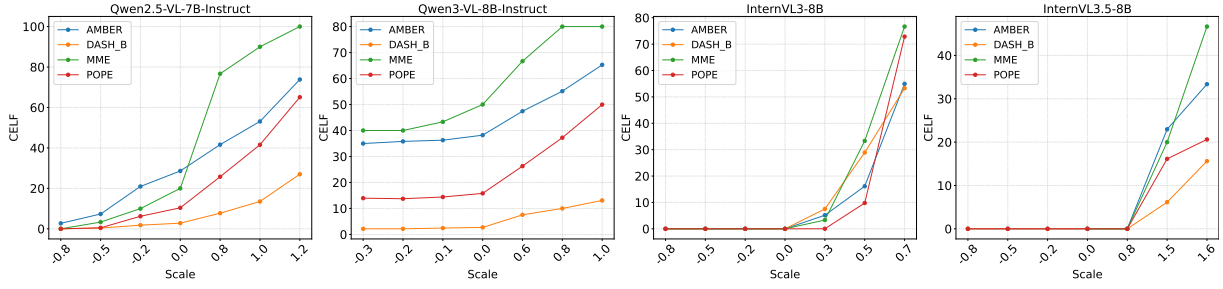


Figure 3: Model performance with different steering coefficients.

This discrepancy is especially pronounced for the InternVL family. Despite strong EA across all four benchmarks, both InternVL3-8B-HF and InternVL3.5-8B-HF obtain zero NLP and zero CELF on every dataset. In other words, whenever these models are asked to localize an absent object, they never abstain, even in cases where they can correctly recognize that the queried object is absent in the existence stage. This is a particularly strong form of existence-localization contradiction.

The Qwen models exhibit slightly better conditional behavior, but the overall trend remains the same. Qwen2.5-VL-7B-Instruct and Qwen3-VL-8B-Instruct achieve non-zero NLP and CELF on all benchmarks, with the best results observed on MME object existence and AMBER. Even so, their CELF scores remain far below their corresponding EA scores. For example, Qwen3-VL-8B-Instruct reaches an EA of 98.3 on MME, but its CELF is only 50.0; similarly, on DASH-B, both Qwen models obtain EA around 74 while CELF remains below 3. These results show that correct absence recognition alone is insufficient to guarantee faithful localization behavior.

Overall, the baseline results support the central claim of this paper: current vision-language models may correctly determine that an object is absent, yet still fail to preserve that judgment when the task is reformulated as localization. This gap between recognition and grounded action is systematic across architectures and benchmarks, and motivates the mitigation approach introduced in the next section.

5.3 Steering Vector Mitigates Existence-Localization Contradiction

As shown in Table 1, steering consistently improves CELF across all model-benchmark pairs, while leaving EA and NLP largely unchanged. For the Qwen models, the gains are substantial. For Qwen2.5-VL-7B-Instruct, CELF increases from

10.4 to 41.6 on POPE, from 28.6 to 51.3 on AMBER, from 20.0 to 90.0 on MME object existence, and from 2.8 to 13.5 on DASH-B. Qwen3-VL-8B-Instruct shows a similar pattern, with CELF improving from 15.8 to 50.0 on POPE, from 38.2 to 65.3 on AMBER, from 50.0 to 80.0 on MME object existence, and from 2.7 to 13.1 on DASH-B. These results indicate that steering makes the Qwen models substantially more likely to preserve a correct absence judgment when subsequently asked to localize the same object.

The effect is especially pronounced for the InternVL models. Both InternVL3-8B and InternVL3.5-8B obtain 0 CELF in the base setting across all benchmarks, indicating a complete lack of conditional abstention behavior: even when the models correctly answer the existence question, they still produce incorrect bounding boxes for non-existent objects. After steering, InternVL3-8B reaches 72.9 CELF on POPE, 55.0 on AMBER, 76.7 on MME object existence, and 53.3 on DASH-B. InternVL3.5-8B also improves from zero to 20.6, 33.4, 46.7, and 15.6 on the same benchmarks, respectively. These improvements show that steering can induce substantial existence-localization faithfulness even in models that initially exhibit no such behavior.

Overall, the consistent CELF improvements show that steering directly addresses the core failure mode targeted by our method: models no longer simply recognize absence in the existence stage, but are also more likely to act consistently with that judgment during localization.

Causal effect of the steering vector. We conduct more in-depth analysis to confirm the role of the steering vector in Section 4.2. Figure 3 reports the CELF metric when applying the steering vector with different coefficients. Overall, we observe a positive correlation between steering coefficients and CELF. Positive steering reliably reduces local-

ization hallucination, and higher coefficients yield better CELF. Similarly, negative steering consistently decreases CELF and makes the model return hallucinated bounding boxes. This result confirms that our steering vector causally controls the ability to recognize an object’s absence under localization prompts.

6 Conclusion

We introduced **Existence–Localization Contradiction**, a failure mode in which vision-language models correctly recognize that a queried object is absent from an image but still produce a non-null localization for that nonexistent referent. To study this gap between absence recognition and grounded action, we proposed a conditional evaluation protocol that pairs existence verification with downstream localization under the same image–query setting. Our experiments show that this contradiction is common even among strong VLMs: models may deny an object’s presence in text while still assigning it a bounding box, sometimes with high confidence. This suggests that localization prompts can weaken or override prior absence judgments, leading to actionable visual hallucinations. At the same time, our intervention results show that lightweight changes can substantially reduce such contradictions while largely preserving localization performance on positive cases. More broadly, our findings underscore the importance of rigorously evaluating hallucinations in general, as well as other model capabilities in particular, rather than relying solely on question-answering performance. Specifically, this need is most salient in task-conditioned settings, where models must translate their perceptual understanding into precise and effective actions. Reliable visual grounding, therefore, requires not only accurate recognition and localization, but also consistency between what a model knows about a scene and the actions it takes. As VLMs are increasingly used in systems where localization outputs guide decisions or physical interactions, preserving absence judgments is essential for trustworthy multimodal deployment.

Limitations

This work focuses on a specific form of grounding inconsistency, where a vision-language model correctly recognizes that a queried object is absent but still produces a non-null localization. As a result, our evaluation does not cover all types

of grounding errors, such as ambiguous references, partial occlusion, attribute-level grounding, relation grounding, or cases where the object is present but visually difficult to distinguish. Our experiments are also limited to a small set of open-source VLMs and object-existence-oriented benchmarks, so the findings may not fully generalize to proprietary models, domain-specific systems, or more complex real-world multimodal interactions. Finally, our mitigation relies on a lightweight steering mechanism that requires validation-based selection of the intervention layer and coefficient; while effective in our setting, it may depend on the model architecture and evaluation distribution, and stronger steering could risk encouraging over-abstention in unseen scenarios.

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A Layer Steering Configuration

This appendix reports the steering scales and layers used for each model in our inference-time intervention. For all models, we steer the same set of intermediate layers, while the steering scales are selected separately for each model family.

Table 2: Steering scales and selected layers used for each model.

Model	Steering scales	Steered layers
Qwen2.5-VL-7B-Instruct	-0.8, -0.5, -0.2, 0.8, 1.0, 1.2	7, 8, 9, 10, 11, 12, 13
Qwen3-VL-8B-Instruct	-0.3, -0.2, -0.1, 0.6, 0.8, 1.0	7, 8, 9, 10, 11, 12, 13
InternVL3-8B	-0.8, -0.5, -0.2, 0.3, 0.5, 0.7	7, 8, 9, 10, 11, 12, 13
InternVL3.5-8B	-0.8, -0.5, -0.2, 0.8, 1.5, 1.6	7, 8, 9, 10, 11, 12, 13

B Prompt Variants

This appendix lists the prompt variants used for localization evaluation. We use model-specific prompt formats to better match the instruction style expected by each model family.

B.1 Qwen Family

For Qwen models, we use natural-language localization prompts. The target object is inserted into the placeholder {object}.

1. Locate a/an {object} in the image and identify its bounding box if it exists.
2. Locate a/an {object} in the image and identify its bounding box.
3. Search the image for a/an {object} and provide its bounding box if it is visible.
4. Detect all instances of {object} in the image and return their locations in the form of coordinates.
5. Identify a/an {object} within the picture and provide its bounding box if present.

B.2 InternVL Family

For InternVL models, we follow the model’s training-style referring-expression prompt format. The target object is inserted into the placeholder {object} and wrapped with <ref> tags.

1. Please provide the bounding box coordinate of the region this sentence describes: <ref>a/an {object}</ref>

2. Please provide the bounding box coordinate of the region this object describes: <ref>a/an {object}</ref>
3. Please provide the bounding box coordinate of the region for this object if it exists: <ref>a/an {object}</ref>
4. Return the bounding box coordinate of the region where this object appears: <ref>a/an {object}</ref>
5. Please provide all bounding box coordinates of the regions where this object is visible: <ref>a/an {object}</ref>

C Ablation on Prompt Variants

We conduct an ablation study to examine whether Existence–Localization Contradiction is sensitive to the wording of the localization prompt. For each model and dataset, we evaluate CELF across multiple prompt variants and report the mean and standard deviation in Table 3. The mean reflects the overall tendency of a model to produce contradictory localization outputs, while the standard deviation captures prompt sensitivity.

The results show that contradiction behavior varies substantially across model families. Qwen-based models exhibit consistently higher CELF scores, especially on AMBER and MME, indicating that they often produce non-null localization outputs even after correctly judging the queried referent to be absent. The relatively large standard deviations for Qwen3-VL-8B-Instruct further suggest that its contradiction rate is sensitive to prompt wording.

In contrast, InternVL models show much lower CELF scores. InternVL3-8B produces near-zero contradictions on POPE and MME, but remains somewhat sensitive on AMBER and DASH-B. InternVL3.5-8B achieves zero CELF across all evaluated datasets and prompt variants, suggesting stronger consistency between absence recognition and localization behavior under this evaluation.

Overall, this ablation shows that Existence–Localization Contradiction is not merely an artifact of a single prompt template. The failure persists across prompt variants for several strong VLMs, while the degree of prompt sensitivity differs substantially across model families.

Table 3: Mean and standard deviation of CELF across different models and datasets.

Model	POPE		AMBER		MME		DASH-B	
	Mean	Std.	Mean	Std.	Mean	Std.	Mean	Std.
Qwen2.5-VL-7B-Instruct	7.72	3.12	23.87	7.28	14.67	7.18	1.45	0.80
Qwen3-VL-8B-Instruct	8.62	8.08	24.41	16.13	34.00	18.55	2.34	1.11
InternVL3-8B	0.00	0.00	1.57	1.83	0.00	0.00	2.19	2.53
InternVL3.5-8B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00